

Examination of Users' Privacy Issues in Live Streaming

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ABSTRACT

Live streaming has become a popular activity worldwide that has warranted research attention on its privacy related issues. The performance-driven, socioeconomic, real-time, on-camera and public nature of live streaming heightens the privacy challenges to users. However, little research has been done to explore the privacy issues in live streaming. Therefore, my research aims to investigate users' privacy concerns and strategies in live streaming, as well as to propose practical design ideas for enhancing privacy management. To achieve these objectives, I conduct three qualitative studies, including interviews with Chinese DouYu streamers (Study 1, published at CSCW), an exploration of streamers' considerations for bystanders' privacy (Study 2, published at CSCW), and an on-going co-design study involving streamers and bystanders (Study 3). The anticipated outcome of this research is to provide valuable design principles and guidelines for improving privacy protection mechanisms on live streaming platforms.

CCS CONCEPTS

• **Security and privacy** → **Human and societal aspects of security and privacy.**

KEYWORDS

privacy, synchronous social media, live streaming, multi-stakeholder, collective privacy management

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1 RESEARCH SITUATION

I began my doctoral journey in fall 2020, under the guidance of Dr. Yao Li in the School of Modeling, Simulation and Training at the University of Central Florida. Currently, I am in my fourth year and plan to graduate in spring 2025. My research aims to better understand users' privacy attitudes and behaviors when they interact with different technologies, and to design better privacy mechanisms to help users control and protect their online privacy. Currently, I am exploring how users' privacy is challenged in the real-time, multimodal, and multi-stakeholder information sharing

setting, such as live streaming. My career aspirations primarily lean toward academia, although I am also open to the possibility of pursuing a career in industry research. I have not had the opportunity to participate in any doctoral consortiums at prior SIGCHI conferences.

2 CONTEXT AND MOTIVATION

Synchronous social media has become a global trend that allows content creators to share their experiences and interact with other users (i.e., viewers) in real-time from any location [16, 17]. A good example of synchronous social media is live streaming, which is my research context. Live streaming is a new form of synchronous social media that enables users to create performance and provide immediate interactions with public audience in real time through webcam and microphone. Due to the improved network bandwidth and widespread high-definition cameras on various devices, live streaming has become increasingly popular. For instance, Twitch is a popular gameplay-based live streaming platform in the US and has approximately 140 million monthly active streamers [19]. DouYu, one of the biggest online live-streaming websites in China, reported 471.8 million registered users in 2022 [18].

The characteristics of live streaming have introduced unanticipated privacy violations that demand further investigation in the following ways. 1) **Live streaming is real-time.** Users in live streaming share their information in real time, thus cannot edit or delete unwanted contents as what users do on asynchronous social media [7]. 2) **Live streaming incorporates multiple modalities of data.** The combination of video and audio modalities can accidentally capture and broadcast users' private moments or conversations without consent, which often raises privacy concerns [11]. 3) **Sharing personal details is key to attract viewers and make revenue.** Streamers are users who broadcast in live streaming. Streamers' information disclosure is part of the content that they create, which is critical for them to draw and retain an audience (e.g., by making viewers feel entertained) and make revenue [8]. Therefore, streamers might struggle between privacy protection and content creation. 4) **Multi stakeholders are involved.** Apart from streamers, bystanders might also be captured when they get close to the streamer's webcam or audio, leading to privacy violations [14]. 5) **The viewers are public and anonymous.** The public and anonymity of viewers make it challenging for users to tailor their information-sharing in time. As a result of characteristics 1-5, there has been increasing news that reports privacy violations in live streaming, such as streamers being harassed and abused by violent stalkers [6] and bystanders being accidentally identified by streamers' audience [1].

However, as HCI researchers have gained more understanding of the live streaming context, few studies have investigated privacy

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issues in such context. Most work related to live streaming studies unique functions of disseminating user-generated content [5], enriching users' self-presentation [10], curating online social interactions [9], fostering online communities [15] and moderating live content [2]. Only a handful of papers have touched on bystanders' privacy concerns in live streaming [3, 11], including my own work [21]. Furthermore, the existing privacy issues in live streaming indicate the insufficient privacy settings of the current live streaming platforms. But few studies have investigated effective design solutions to address user's privacy issues in live streaming. Therefore, it is necessary to untangle users' privacy concerns and management in live streaming and explore practical design solutions to solve the users' privacy issues and coordinate different stakeholders' privacy preferences in live streaming.

3 RESEARCH GOALS

To address the above research gap, my doctoral research aims to answer the following three research questions:

RQ1: What privacy concerns do users (e.g., streamers and bystanders) have in live streaming?

RQ2: What are the coping strategies that users adopt to manage their information disclosure in live streaming?

RQ3: What design ideas are anticipated by users to protect their privacy in live streaming?

4 WORK TO DATE

My research is an ongoing effort aimed at addressing all the RQs. My first two complete papers [20, 21] mainly address RQ1 & RQ2 by examining different stakeholder's privacy through two qualitative studies. Currently, I am in the process of conducting a co-design study for my third paper to answer RQ3. In the following subsections, I will present the findings of my work in more detail.

4.1 Study 1: Streamers' Privacy Concerns and Strategies in Live Streaming Information Disclosure (RQ1&RQ2)

4.1.1 The rationale to study streamers' privacy concerns. Prior work shows that streamers are prone to present attractive performances to their audience, which could generate financial revenues for streamers [4]. Thus, streamers adopt various self-presentation strategies so that their performance attracts, satisfies, and entertains the audience. However, little attention has been focused on streamers' privacy concerns and practices in live performance creation. As streamers disclose large amounts of personal information in producing live performances, their privacy concerns and strategies might interplay with their desire to generate an attractive performance and to seek financial revenues. They might encounter tensions between content creation and privacy protection. Additionally, while viewers mostly remain anonymous, streamers share a great deal of identifiable information in live streaming, such as their physical appearance and physical locations, which could potentially cause privacy risks, such as boundary disruption, stalking and harassment. Thus, I conduct this study to investigate the privacy issues in streamers' content creation.

4.1.2 Method. In this study, I conducted 20 semi-structured interviews with Chinese DouYu streamers through voice chat in October 2020. The participants were recruited through direct contact and snowball sampling. I carefully considered full-time/part-time, length of streaming and number of viewers when recruiting participants because I wanted to explore privacy concerns and strategies for different types of streamers. Each interviewee was compensated 70 (about 10 dollars). Thematic analysis was applied to analyze the data.

4.1.3 Findings. We found two categories of privacy concerns that participants had during live streaming. First, there are tensions between streamers' privacy and attractiveness. Although streamers had concerns about sharing their physical features, such as look and apparel, during live streaming, they still chose to disclose such information to attract audiences. There was also some information that streamers chose not to disclose, such as their romantic relationship. Because they had concerns that disclosing such information would harm their attractiveness and lose their audience. Second, there are tensions between streamers' privacy and interpersonal boundaries with the audience. Streamers were unwilling to disclose their mailing address, location, email address and game ID to avoid physical or virtual contact with their audience. Because they were worried about being harassed by vicious viewers online or in real world. To address the privacy concerns, streamers would adopt strategies such as using a green curtain for a better environment or disabling notifications to avoid pop messages during streaming. However, these strategies are not perfect due to the unpredictability and uncertainty of live streaming and might add cognitive burdens to streamers. As such, we propose to integrate intelligent webcam filters to hide streamers' unwanted information without harming their attractiveness and also insert intelligent and granular notifications to help streamers cope with the change of context in live streaming. This original study was published in the proceedings of the ACM on Human-Computer Interaction [20].

4.2 Study 2: Live Streamers' Considerations and Strategies for Bystander's Privacy Management (RQ1&RQ2)

4.2.1 The rationale to study streamers' considerations towards bystander's privacy. The majority of privacy research in live streaming is to understand bystanders' privacy issues from the bystanders' perspective [3, 14]. No prior study pays attention to how streamers perceive their bystander's information disclosure and how they protect their bystander's privacy from being violated. It is important to examine streamers' considerations towards bystanders' privacy because bystanders have to rely on the streamers to protect their privacy. Prior work has pointed out that bystanders have little agency over their privacy, whereas streamers usually have direct control over the video and audio sharing [12, 13]. Many popular streaming platforms, such as Twitch and DouYu, do not provide any mechanisms to let bystanders directly manage their privacy, nor allow bystanders to request streamers to protect their privacy. Thus, only when streamers are self-motivated to raise awareness and take actions can bystanders' privacy be protected.

4.2.2 Method. In this study, I interviewed 25 Chinese DouYu streamers through voice chat to understand their considerations and practices regarding bystanders' privacy in live streaming. The participants included 15 males and 10 females. They were recruited through direct contact and word-of-mouth. Since the focus of this study is on streamers' attitudes to their bystanders' privacy, I intentionally recruited streamers with various occupational backgrounds, lengths of streaming, and numbers of viewers to have a better understanding of diverse streamers' perspectives. I paid 70 CNY (about 10 dollars) for each participant. The data was analyzed through thematic analysis.

4.2.3 Findings. We found that although streamers cared about their bystanders' privacy, they made privacy decisions largely based on their own assumptions through perceived information sensitivity, perceived bystander's personality and perceived reactions from bystander. For example, streamers believe extroverted bystanders are more willing to share about themselves, so streamers feel it is fine to share extroverted bystanders' information with the audience. Streamers also believe bystander's information such as voice is not sensitive so they will not avoid sharing such information in their streaming. However, it is unknown if bystanders truly agree with these assumptions, since our streamers did not check with their bystanders about their considerations.

To protect bystander's privacy, streamers were willing to adopt strategies, such as disabling video or audio feeds to avoid disclosing bystander's physical appearance and conversation or scheduling streaming time when bystanders were not at home. However, these strategies may not always be effective given the lack of features on current streaming platforms that facilitate effective real-time communication between streamers and bystanders about privacy preferences.

Our findings highlight the importance of having better two-way communication between streamers and bystanders about bystander's privacy needs. We suggest sending notifications to streamers to arouse bystander's awareness of bystander's involvement before streaming. Unlike other contexts such as group photo sharing, recognizing changes in context during live streaming can be challenging for both streamers and bystanders. Many streamers live with their bystanders, making it easy to forget they are streaming and engage in personal conversations. Additionally, bystanders often appear suddenly, leaving streamers with little time to adjust their information sharing practices. So we suggest having a proximity sensor to alert streamers about the incoming bystanders in advance, and intelligent techniques to block bystanders and change bystanders' voice. This original study was accepted in the proceedings of the ACM on Human-Computer Interaction [21].

4.3 Study 3: Design Privacy Settings in Live Streaming (In Process, RQ3)

4.3.1 The rationale to conduct a co-design study. The results of the two preliminary studies indicate that streamers and bystanders have privacy concerns about disclosing their personal information in live streaming. However, current streaming platforms fail to provide sufficient privacy protection features to protect streamers' privacy and support streamers and bystanders to co-manage bystanders' privacy. Therefore, this study aims to improve the privacy design

of live streaming tools so that streamers and bystanders feel more secure about their personal information.

4.3.2 The progress of the study. This study is still in process. So far, I have completed the data collection and am focusing on the data analysis. We have conducted 11 codesign sessions involving 26 streamers and bystanders to explore their privacy concerns, their artifacts and the rationales between each design idea. I am working on analyzing the data through thematic analysis.

4.3.3 Expected outcome. The expected outcome of this study is to enrich our understanding of streamers' and bystanders' needs and their expectations in privacy mechanisms for live streaming. The participants will provide a set of design factors that they consider in their desirable privacy mechanisms to address their concerns. The factors may be streamer/bystander-centric mechanisms or cooperative mechanisms. The findings may also highlight the tension of privacy expectations between the streamers and bystanders in live streaming. Based on the participants' conceptualized designs, a number of concrete design suggestions will be made to better support and balance different stakeholders' privacy needs in live streaming.

5 FUTURE DISSERTATION WORK

Following this co-design study, my plan is to implement a series of prototypes of privacy systems based on the design ideas provided by streamers and bystanders and then conduct tests to evaluate the effectiveness of these design systems in safeguarding the privacy of both streamers and bystanders. This evaluation process will enable me to test whether the design systems effectively address the privacy concerns of both stakeholders and identify areas that may require further improvement. Additionally, our findings indicate that marginalized groups such as women users and children are particularly vulnerable in the live streaming environment. Therefore, I plan to conduct further research to address and mitigate the unique challenges and privacy concerns marginalized groups face. For example, I intend to conduct a qualitative study aimed at understanding the privacy concerns specific to female streamers.

6 CONTRIBUTION

The contributions of my doctoral research are as follows:

First, my research provides an empirical account of privacy challenges in a novel and understudied context – live streaming. Most social media examined are asynchronous, which means that the content production and consumption are in sequence rather than in real-time. Such findings may not apply to privacy management on synchronous social media such as live streaming platforms where information sharing and consuming are in real-time. Thus, my work aims to extend HCI research on social media privacy by untangling the privacy concerns and management of users in real time context and exploring design implications to support their privacy management.

Second, my work uncovers the intertwined relationships among privacy management, content creation, and revenue generation that do not exist in previously studied social media. Unlike the social media platforms that are used to maintain social network (e.g., Facebook), live streaming is for users to share attractive performances

to make financial revenues. Thus, my research demonstrates that the privacy concerns of users on platforms with socioeconomic implications extend beyond personal information exposure. These concerns are intricately linked to whether sharing information enhances or undermines their performance and attractiveness to the public audience, ultimately impacting their financial revenue.

Third, my research highlights the importance of collective privacy management within synchronous social media regarding the multi-stakeholders nature of live streaming. While prior studies in such context have primarily focused on addressing privacy concerns at an individual stakeholder level, for example, investigating bystander's privacy concerns from bystander's perspective, my research underscores the essential role of collective efforts in ensuring the privacy of both streamers and bystanders.

Fourth, my research proposes a set of practical design patterns and guidelines to safeguard privacy for various stakeholders in synchronous social media. These design implications are not limited to live streaming but also provide privacy practices in other synchronous social media.

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